Sequence and Series Question Bank

1. Test the convergence of the series: $\frac{1}{1+x}+\frac{1}{2+x}+\frac{1}{3+x}+\cdots$
2. Show that the following series is divergent: $\sum_{n=1}^{\infty}\sin\left(\frac{1}{n}\right)$
3. Test the convergence of the series: $\frac{1}{1\cdot2\cdot3}+\frac{3}{2\cdot3\cdot4}+\frac{5}{3\cdot4\cdot5}+\cdots\infty$
4. Test for convergence: $\sum_{n=1}^{\infty}\left[\sqrt[3]{n^3+1}-n\right]$
5. Show that the series: $\frac{2}{1^p}+\frac{3}{2^p}+\frac{4}{3^p}+\cdots$ is **convergent** for $p>2$ and **divergent** for $p\leq 2$.
6. Test for convergence: $\sum_{n=1}^{\infty}\left(\frac{2^n+3}{3^n+1}\right)^{1/2}$
7. Test for the convergence of the series: $\frac{1}{4\cdot7\cdot10}+\frac{4}{7\cdot10\cdot13}+\frac{9}{10\cdot13\cdot16}+\cdots\infty$
8. Test for convergence: $\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}\tan\left(\frac{1}{n}\right)$.
9. Test the convergence of the series: $\sum_{n=1}^{\infty}\frac{x^n}{n^r+1}$
10. Test the convergence of the series: $\sum_{n=1}^{\infty}\frac{(n+1)x^n}{n^3}\quad(x>0)$.
11. Test the convergence of the series: $\sum_{n=1}^{\infty}\left(\frac{n^2}{2^n}+\frac{1}{n^2}\right)$
12. Test for the convergence: $\sum_{n=1}^{\infty}\frac{1\cdot3\cdot5\cdots(2n-1)}{2\cdot4\cdot6\cdots(2n)}x^{n-1}\quad(x>0)$.
13. Test for the convergence of: $1+\frac{2}{5}x+\frac{6}{9}x^2+\cdots+\left(\frac{2^n-2}{2^n+1}\right)x^{n-1}+\cdots\quad(x>0)$.
14. Discuss the nature of the series: $\frac{2}{3\cdot4}x+\frac{3}{4\cdot5}x^2+\frac{4}{5\cdot6}x^3+\cdots\infty\quad(x>0)$.
15. Test the convergence of the series: $1+\frac{x}{2}+\frac{x^2}{5}+\frac{x^3}{10}+\cdots$
16. Test for convergence of the series: $x+\frac{1}{2}\cdot\frac{x^3}{3}+\frac{1\cdot3}{2\cdot4}\cdot\frac{x^5}{5}+\frac{1\cdot3\cdot5}{2\cdot4\cdot6}\cdot\frac{x^7}{7}+\cdots$
17. Test the convergence of the series: $1+\frac{3}{7}x+\frac{3\cdot6}{7\cdot10}x^2+\frac{3\cdot6\cdot9}{7\cdot10\cdot13}x^3+\cdots\quad(x>0)$.
18. Find the nature of the series: $\sum_{n=1}^{\infty}\frac{(n!)^2}{(2n)!}x^n\quad(x>0)$.
19. Test for convergence of the series whose n^{th} term is:

    $(i)\quad a_n=\frac{1}{\left(1+\frac{1}{n}\right)^{n^2}}\quad(ii)\quad a_n=\frac{1}{3^n-1}$
20. Test the convergence of the series: $\sum_{n=1}^{\infty}\frac{[(n+1)x]^n}{n^{n+1}}$.
21. Show that the series: $\left[\frac{2^2}{1^2}-\frac{2}{1}\right]^{-1}+\left[\frac{3^3}{2^3}-\frac{3}{2}\right]^{-2}+\left[\frac{4^4}{3^4}-\frac{4}{3}\right]^{-3}+\cdots$ is **convergent**.